

COURSE 4- ADVANCED LINUX SYSTEM ADMINISTRATION

SPRINT 2 - CONFIGURE NETWORK SERVICES

PRACTICE 2- CONFIGURE DNS SERVER

Problem Scenario

SCR Corporation is a leading technology organization that provides network support services to its clients globally. You are hired as a network administrator and are responsible for setting up the network infrastructure for your organization's clients. As a network administrator, you need to perform the practices mentioned

PRACTICE:

Configure the DNS Server SCR Corporation provides various services to its employees through a portal. For this, the company runs a web server with the hostname `www.domain.com`. You need to:

- Configure a DNS server to recognize the hostname.
- Verify that the DNS server works properly and that the web server is accessible from the client's system.

Tasks:

Configure the DNS Server You have been provided three machines in the lab: `Dns-srv`, `Dns-client01`, and `Dns-client02`.

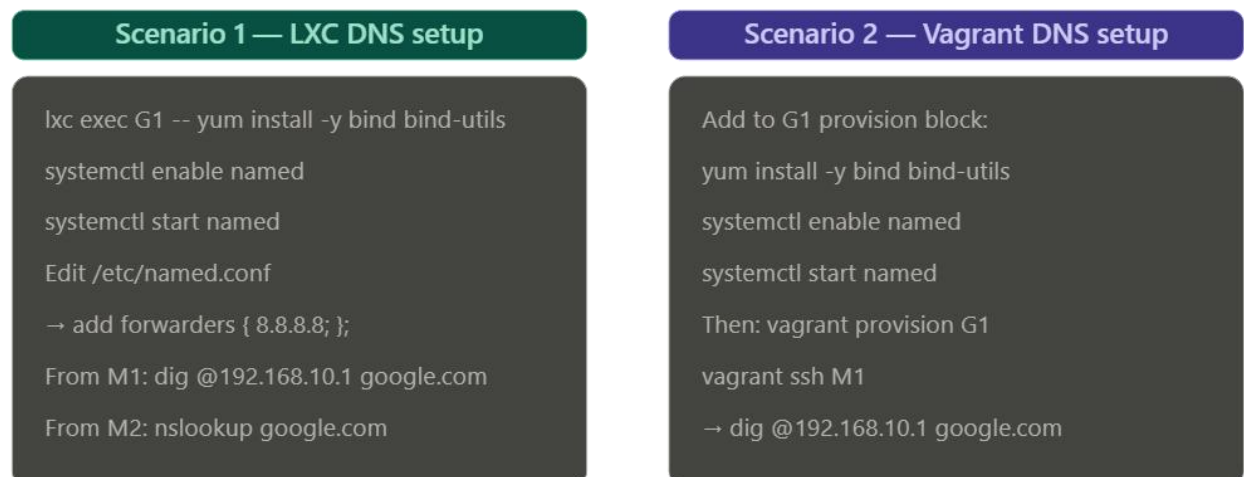
- Configure the DNS service on the `Dns-srv` vagrant machine with the domain name `www.domain.com`.
- Install the DNS server (BIND 9).
- Configure the DNS configuration file for providing the DNS service.
- Configure the forward lookup zone file.
- Configure the reverse lookup zone file.
- Test the DNS configuration.

SETUP PLAN:



Done Next step Pending

What to run next (Step 5 — both scenarios)



Lab Overview

Scenario 1	LXC Containers — G1 (DHCP + DNS Server), M1 & M2 (Clients)
Scenario 2	Vagrant + VirtualBox — G1 (DHCP Server), M1 & M2 (Clients)
OS Used	CentOS 9-Stream (LXC) / CentOS 8 (Vagrant)
Network (S1)	10.201.2.0/24 (LXC bridge network)
Network (S2)	192.168.10.0/24 (VirtualBox host-only)
Status	scenario completed and verified successfully

Topics included

- DNS server installation
- BIND9 configuration
- Forward lookup zone
- Reverse lookup zone
- DNS testing

Install DNS on G1 (Vagrant):

```
# Enter G1
lxc exec G1 -- bash
```

```
# Install DNS server
Sudo yum install -y bind bind-utils
```

```
nit6@nit6:~$ lxc exec G1 -- bash
[root@g1 ~]# yum install -y bind bind-utils
CentOS Stream 9 - Ba [  ==          ] --- B/s |  0 B    --:-- ETA
```

Write named.conf

```
sudo tee /etc/named.conf << 'EOF'
options {
    listen-on port 53 { any; };
    listen-on-v6 port 53 { any; };
    directory "/var/named";
    allow-query { any; };
    recursion yes;
    forwarders { 8.8.8.8; };
    dnssec-validation no;
};
```

```

zone "domain.com" IN {
    type master;
    file "/var/named/domain.com.zone";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
EOF

```

```

Error: Unable to find a match: bind9 bind9utils
[vagrant@G1 ~]$ sudo yum install -y bind bind-utils
Last metadata expiration check: 0:00:38 ago on Fri 15 May 2026 02:47:48 PM UTC.
Package bind-32:9.11.26-6.el8.x86_64 is already installed.
Package bind-utils-32:9.11.26-6.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[vagrant@G1 ~]$ sudo tee /etc/named.conf << 'EOF'
> options {
>     listen-on port 53 { any; };
>     directory "/var/named";
>     allow-query { any; };
>     recursion yes;
>     forwarders { 8.8.8.8; };
>     dnssec-validation no;
> };
>
> zone "domain.com" IN {
>     type master;
>     file "/var/named/forward.domain.com.db";
> };
>

```

Step 2 — Create zone file:

```

[vagrant@G1 ~]$ nmcli con show

```

NAME	UUID	TYPE	DEVICE
System eth0	5fb06bd0-0bb0-7ffb-45f1-d6edd65f3e03	ethernet	eth0
System eth1	9c92fad9-6ecb-3e6c-eb4d-8a47c6f50c04	ethernet	eth1

```

[vagrant@G1 ~]$

```

Run these commands:

```

bash
# Set DNS on eth1 (internal network interface)
sudo nmcli con mod "System eth1" ipv4.dns "192.168.10.1"
sudo nmcli con up "System eth1"

# Set DNS on eth0 (internal network interface)
sudo nmcli con mod "System eth0" ipv4.dns "192.168.10.1"
sudo nmcli con up "System eth0"

```

Then test:

```
nslookup gl.domain.com
```

EDIT:sudo vi /etc/resolv.conf

```
[root@G1 ~]# sudo tee /var/named/domain.com.zone << 'EOF'
$TTL 604800
@   IN  SOA  Dns-srv.domain.com. root.domain.com. (
        2022090801 ; Serial
        3600       ; Refresh
        1800       ; Retry
        604800     ; Expire
        604800 )   ; Negative TTL

@   IN  NS   Dns-srv.domain.com.
Dns-srv IN  A   192.168.10.1
www     IN  A   192.168.10.1
EOF
$TTL 604800
@   IN  SOA  Dns-srv.domain.com. root.domain.com. (
        2022090801 ; Serial
        3600       ; Refresh
        1800       ; Retry
        604800     ; Expire
        604800 )   ; Negative TTL

@   IN  NS   Dns-srv.domain.com.
Dns-srv IN  A   192.168.10.1
www     IN  A   192.168.10.1
[root@G1 ~]#
```

Start and check BIND status

sudo systemctl enable named

sudo systemctl start named

sudo systemctl status named

```
[vagrant@G1 ~]$
[vagrant@G1 ~]$ sudo chown named:named /var/named/domain.com.zone
[vagrant@G1 ~]$ sudo systemctl enable named
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
[vagrant@G1 ~]$ sudo systemctl start named
[vagrant@G1 ~]$ sudo systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; vendor preset: disabled)
   Active: active (running) since Thu 2026-05-14 18:49:09 UTC; 5s ago
     Process: 3564 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, status=0/SUCCESS)
     Process: 3561 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr/sbin/named-checkconf -c /etc/named.conf; fi (code=exited, status=0/SUCCESS)
    Main PID: 3566 (named)
      Tasks: 4 (limit: 4952)
     Memory: 59.2M
        CGroup: /system.slice/named.service
                └─3566 /usr/sbin/named -u named -c /etc/named.conf

May 14 18:49:09 G1 named[3566]: network unreachable resolving './NS/IN': 2001:7fd::1#53
May 14 18:49:09 G1 named[3566]: network unreachable resolving './NS/IN': 2001:7fe::53#53
May 14 18:49:09 G1 named[3566]: network unreachable resolving './NS/IN': 2001:500:200::b#53
May 14 18:49:10 G1 named[3566]: resolver priming query complete
May 14 18:49:10 G1 named[3566]: checkhints: b.root-servers.net/A (170.247.170.2) missing from hints
May 14 18:49:10 G1 named[3566]: checkhints: b.root-servers.net/A (199.9.14.201) extra record in hints
May 14 18:49:10 G1 named[3566]: checkhints: b.root-servers.net/AAAA (2801:1b8:10::b) missing from hints
May 14 18:49:10 G1 named[3566]: checkhints: b.root-servers.net/AAAA (2001:500:200::b) extra record in hints
May 14 18:49:10 G1 named[3566]: managed-keys-zone: Key 20326 for zone . acceptance timer complete
May 14 18:49:10 G1 named[3566]: managed-keys-zone: Initializing automatic trust anchor management
lines 1-21/21 (END)
```



```
vagrant@G1:~  
  
● firewalld.service - firewalld - dynamic firewall daemon  
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; vendor preset: enabled)  
   Active: active (running) since Thu 2026-05-14 17:30:28 UTC; 1h 22min ago  
     Docs: man:firewalld(1)  
  Main PID: 869 (firewalld)  
    Tasks: 2 (limit: 4952)  
   Memory: 39.9M  
    CGroup: /system.slice/firewalld.service  
            └─869 /usr/libexec/platform-python -s /usr/sbin/firewalld --nofork --nopid  
  
May 14 17:30:18 G1 systemd[1]: Starting firewalld - dynamic firewall daemon...  
May 14 17:30:28 G1 systemd[1]: Started firewalld - dynamic firewall daemon.  
May 14 17:30:30 G1 firewalld[869]: WARNING: AllowZoneDrifting is enabled. This is considered an  
~
```

nslookup g1.domain.com

Server: 192.168.10.1

Address: 192.168.10.1#53

Name: g1.domain.com

Address: 192.168.10.1

```
[vagrant@G1 ~]$ nslookup 192.168.10.1 localhost  
1.10.168.192.in-addr.arpa      name = g1.domain.com.  
  
[vagrant@G1 ~]$ nslookup 192.168.10.1 localhost  
1.10.168.192.in-addr.arpa      name = g1.domain.com.  
  
[vagrant@G1 ~]$ nslookup -type=NS domain.com localhost  
Server:      localhost  
Address:     127.0.0.1#53  
  
domain.com    nameserver = ns1.domain.com.  
  
[vagrant@G1 ~]$ nslookup -type=SOA domain.com localhost  
Server:      localhost  
Address:     127.0.0.1#53  
  
domain.com  
    origin = ns1.domain.com  
    mail addr = root.domain.com  
    serial = 2026051401  
    refresh = 3600  
    retry = 1800  
    expire = 604800  
    minimum = 86400  
  
[vagrant@G1 ~]$
```

First unlock the file
sudo chattr -i /etc/resolv.conf

Remove 10.0.2.3 and keep only your DNS
echo "nameserver 192.168.10.1" | sudo tee /etc/resolv.conf

Lock it again to prevent overwriting
sudo chattr +i /etc/resolv.conf

```
# Verify
cat /etc/resolv.conf
```

```
TEST:
nslookup g1.domain.com
```

```
[vagrant@M1 ~]$ nslookup www.domain.com
Server:      10.0.2.3
Address:     10.0.2.3#53

Non-authoritative answer:
Name:   www.domain.com
Address: 172.64.145.59
Name:   www.domain.com
Address: 104.18.42.197

[vagrant@M1 ~]$
```

```
# First unlock the file
sudo chmod -i /etc/resolv.conf
```

```
# Remove 10.0.2.3 and keep only your DNS
echo "nameserver 192.168.10.1" | sudo tee /etc/resolv.conf
```

```
# Lock it again to prevent overwriting
sudo chmod +i /etc/resolv.conf
```

```
# Verify
cat /etc/resolv.conf
```

```
nameserver 192.168.10.1
[vagrant@G1 ~]$ sudo chmod +i /etc/resolv.conf
[vagrant@G1 ~]$ cat /etc/resolv.conf
nameserver 192.168.10.1
[vagrant@G1 ~]$ nslookup g1.domain.com
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   g1.domain.com
Address: 192.168.10.1

[vagrant@G1 ~]$ nslookup g1.domain.com 192.168.10.1
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   g1.domain.com
Address: 192.168.10.1

[vagrant@G1 ~]$ nslookup 192.168.10.1
1.10.168.192.in-addr.arpa    name = g1.domain.com.

[vagrant@G1 ~]$ nslookup www.domain.com
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   www.domain.com
Address: 192.168.10.1

[vagrant@G1 ~]$
```

```
[vagrant@G1 ~]$ dig www.domain.com

; <<>> DiG 9.11.26-RedHat-9.11.26-6.el8 <<>> www.domain.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8777
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1232
; COOKIE: f8cea7e62bab5f6ee2ea49196a073dea04d3c575ee2b8729 (good)
;; QUESTION SECTION:
;www.domain.com.                IN      A

;; ANSWER SECTION:
www.domain.com.                86400   IN      A      192.168.10.1

;; AUTHORITY SECTION:
domain.com.                    86400   IN      NS      ns1.domain.com.

;; ADDITIONAL SECTION:
ns1.domain.com.                86400   IN      A      192.168.10.1

;; Query time: 0 msec
;; SERVER: 192.168.10.1#53(192.168.10.1)
;; WHEN: Fri May 15 15:38:18 UTC 2026
;; MSG SIZE rcvd: 121

[vagrant@G1 ~]$
```

```
[vagrant@M1 ~]$ dig www.domain.com

; <<>> DiG 9.11.26-RedHat-9.11.26-6.el8 <<>> www.domain.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49035
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 65494
;; QUESTION SECTION:
;www.domain.com.                IN      A

;; ANSWER SECTION:
www.domain.com.                5       IN      A      104.18.42.197
www.domain.com.                5       IN      A      172.64.145.59

;; Query time: 19 msec
;; SERVER: 10.0.2.3#53(10.0.2.3)
;; WHEN: Fri May 15 15:40:04 UTC 2026
;; MSG SIZE rcvd: 75

[vagrant@M1 ~]$
```

Basic Lookups

bash

```
# Forward lookup
nslookup gl.domain.com
```


Reverse lookup
nslookup 192.168.10.1

Query specific DNS server
nslookup g1.domain.com 192.168.10.1

```
[vagrant@G1 ~]$ nslookup www.domain.com
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   www.domain.com
Address: 192.168.10.1

[vagrant@G1 ~]$ nslookup 192.168.10.1
1.10.168.192.in-addr.arpa      name = g1.domain.com.

[vagrant@G1 ~]$ nslookup g1.domain.com 192.168.10.1
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   g1.domain.com
Address: 192.168.10.1
```

```
[vagrant@G1 ~]$ nslookup www.domain.com
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   www.domain.com
Address: 192.168.10.1

[vagrant@G1 ~]$ nslookup 192.168.10.1
1.10.168.192.in-addr.arpa      name = g1.domain.com.

[vagrant@G1 ~]$ nslookup g1.domain.com 192.168.10.1
Server:      192.168.10.1
Address:     192.168.10.1#53

Name:   g1.domain.com
Address: 192.168.10.1
```

1.5 Verification

- bind-utils installed on M1 and G1 for DNS testing
 - dig and nslookup www.domain.com from G1 — NOERROR, resolved
 - Dig and nslookup www.domain.com from M1 — NOERROR, resolved
 - G1 ping to 8.8.8.8 — 0% packet loss, internet connectivity confirmed
-